

**Mat2045 - Numerical
Methods for Engineers****COURSE SYLLABUS****1. Welcome**

Welcome to Numerical Methods for Engineers. In this course, you will develop an intuitive and practical understanding of numerical methods for solving mathematical problems, including error analysis and computational problem solving. You will also implement and test numerical methods using MATLAB. Regular attendance, preparation, and practice with self-study exercises will support your progress throughout the semester.

2. Instructor Information

Instructor	Lavdie Rada Ülgen Problem-solving classes: Research Assistant
Office & Office Hours	Lavdie Rada Ülgen: D310; Monday: 12:30-14:30, Tuesday 12:30-14:30
E-mail	lavdie.rada@bau.edu.tr
CV link	https://scholar.google.com/citations?user=wRqB-1EAAAAJ&hl=tr

3. Course Overview

Semester	Fall 2026-27
Credits/ ECTS	4 / 5
Classroom & Time	Monday 15:30-18:20 (Section#4) - A204 Thursday 15:30-16:20 (Section#1) – D405 Wednesday 12:30-15:20 (Section#2) – D404 Thursday 15:30-18:20 (Section#3) – U305
Delivery Mode	☒ Face-to-Face ☐ Fully Online - Synchronous ☐ Fully Online - Asynchronous
Course Type	☒ Compulsory ☐ Elective
Prerequisite	Students taking this course must know Calculus and Linear Algebra.

4. Course

Course Objectives

This course is concerned with the mathematical derivation, description, and analysis of obtaining numerical solutions of mathematical problems. Students should obtain an intuitive and working understanding of some numerical methods for the basic problems of numerical analysis. They should gain some appreciation of the concept of error and of the need to analyze and predict it. Topics cover linear and nonlinear systems of equations, interpolation, curve fitting using the least-squares method, numerical differentiation and integration, discrete Fourier transformation, power method for eigenvalues and eigenvectors of matrix, singular value decomposition. Students should gain some experience in the implementation of numerical methods by using MATLAB.

Course Learning Outcomes (CLOs)

1. Define errors, big O notation, use Taylor's theorem.
2. Solve nonlinear algebraic equations.
3. Solve linear systems and to use iterative methods for linear systems.
4. Solve systems of nonlinear algebraic equations.
5. Use interpolation methods and polynomial approximation for a given data, piecewise linear interpolation and spline function interpolation.
6. Use least-squares method for curve fitting.
7. Approximate the dominant eigenvalue and corresponding eigenvector of the matrix and to calculate singular value decomposition of a matrix and apply it on image processing.
8. Approximate derivatives and integrals numerically and calculate discrete Fourier transformation.
9. Implement numerical methods on MATLAB and test their programs behavior through expected results in accordance with the Numerical Analysis theory.

Course Alignment with the Sustainable Development Goals (SDGs)

☐ SDG 1 No Poverty	☐ SDG 2 Zero Hunger	☐ SDG 3 Good Health and Well-Being	☒ SDG 4 Quality Education	☐ SDG 5 Gender Equality	☐ SDG 6 Clean Water and Sanitation	☐ SDG 7 Affordable and Clean Energy	☐ SDG 8 Decent Work and Economic Growth	☐ SDG 9 Industry, Innovation and Infrastructure
☐ SDG 10 Reduced Inequalities	☐ SDG 11 Sustainable Cities and Communities	☐ SDG 12 Responsible Consumption and Production	☐ SDG 13 Climate Action	☐ SDG 14 Life Below Water	☐ SDG 15 Life on Land	☐ SDG 16 Peace, Justice and Strong Institutions	☐ SDG 17 Partnerships for the Goals	☐ This course is not directly related to any SDG.

5. Learning Experience

Instructional Strategies (Macro-level course design)	Teaching Methods (Main modes of instruction)	Teaching Techniques (Interaction and engagement tools)	Learning Activities (Student workload / ECTS-based tasks)	Learning Technologies (Digital tools & platforms)
☒ Expository Teaching ☐ Inquiry-Based ☐ Problem-Based ☐ Project-Based ☐ Research-Based ☐ Experiential ☐ Cooperative ☐ Other	☒ Lecture ☐ Discussion ☐ Case Study ☒ Problem Solving ☐ Experiment ☐ Simulation ☐ Field Study/Practicum ☒ Other: MATLAB implementation	☐ Questioning ☐ Brainstorming ☐ Group Work ☐ Role-Playing ☐ Expert Talk ☐ Panel / Debate ☐ Field Trip ☐ Other	☐ Group/Team Work ☒ Reading, Watching, Listening ☐ Observation ☐ Presentation ☐ Writing/Reflection ☐ Self/Peer Evaluation ☐ Internship ☐ Educational Game ☒ Other: Self-study exercises	☒ MS Teams ☐ Simulation Software ☐ AI Tools ☐ Web 2.0 Tools ☐ Virtual Lab ☒ Videos ☐ Podcasts ☒ Other: MATLAB

6. CLO, Assessment and AI Use

CLO No.	Assessment Component	Weight (%)	AI Use Level of the student *
-	Assignment/Project (if assigned via Teams)	-	☐ AI Not Permitted ☐ Limited and Conditional Use ☐ Broad Use
1-5	Midterm Exam	35	☒ AI Not Permitted ☐ Limited and Conditional Use ☐ Broad Use
2-4, 7-9	Quizzes I-IV (4%, 4%, 4%, 8%)	20	☒ AI Not Permitted ☐ Limited and Conditional Use ☐ Broad Use
1-12	Final Exam	45	☒ AI Not Permitted ☐ Limited and Conditional Use ☐ Broad Use

*The permitted use of AI for each assessment task will be specified in the relevant task instructions.

7. Rubrics

No separate non-exam assignment/project is included in the current grading structure. Assessment announcements and course materials will be shared via Microsoft Teams.

8. Generative AI Policy

Bahçeşehir University AI Use Policy

In courses, you may use Microsoft Copilot and other generative AI tools to support your learning, in accordance with the guidelines set by your instructor. However, AI tools do not replace your responsibility for original thinking, analysis, interpretation, or academic work. You remain fully responsible for the academic integrity and originality of everything you submit. When using AI tools, you must comply with Bahçeşehir University's Artificial Intelligence Regulation, Academic Integrity Principles, regulations on the protection of personal data, and any course-specific requirements.

BAU Artificial Intelligence Regulation: <https://ai.bau.edu.tr/en/institutional/principles-policies/yz-mevzuati/>

Student AI Declaration (for each assignment)

If AI use is permitted or required for an assignment, you are expected to submit an AI Use Declaration Form with each assignment. If AI use is not permitted, do not use AI tools and do not submit a declaration.

STUDENT AI DECLARATION FORM

AI Tool(s) used	
Purpose of use	
Review and responsibility	I declare that I have reviewed the content generated by artificial intelligence, made the necessary revisions, and am responsible for the accuracy and academic integrity of the submitted work.
Student's full name	
Date and signature	

Citing AI Use in Your Work

When AI tools contribute to submitted work, you are expected to acknowledge this use clearly and cite the tool in accordance with the citation style **APA (7th ed.)**.

9. Academic Integrity

Academic integrity is a fundamental expectation in this course. You are responsible for ensuring that all work you submit is your own and complies with Bahçeşehir University's Academic Integrity Principles and course requirements. If there is a reasonable concern that you may have engaged in cheating, attempted cheating, plagiarism, unauthorized use of AI, or any other form of academic misconduct in an examination, assignment, project, or other assessment, the matter will be reviewed in accordance with Law No. 2547 and the relevant Bahçeşehir University regulations.

10. Course Policies

Communication

Use e-mail for questions outside class hours and follow announcements published on Microsoft Teams. Course announcements and materials will be posted on the Teams course page. All announcements will be made in Türkiye time.

Attendance

Attendance is governed by Engineering Faculty attendance regulations. Each student must attend a minimum of 60% of the classes. The BAU Senate will decide the NA rule at the end of the semester.

Digital Devices

Mobile phones, tablets, and laptop computers may be used only for teaching purposes. During class, mobile phones should be on silent and kept out of sight. In important health, family, or personal situations, you may use vibration mode; if you need to make a call, leave the classroom and return afterward.

Respectful Learning Environment

As future professional engineers, students are expected to communicate respectfully and cordially with classmates and the course instructors and to participate positively in classroom activities and discussions.

Accessibility

At Bahçeşehir University, we care about ensuring that every student can participate in classes equally and comfortably. If you have any disability or specific learning need, we kindly ask you to contact the Bahçeşehir University Disabled Student Unit (<https://bau.edu.tr/content/4321-disabled-student-unit>) at the beginning of the semester so that we can provide you with the most appropriate support. Once you have shared your needs with the relevant unit, please inform me as well so that we can arrange the most suitable accommodations for you.

Privacy & Copyright

Taking photos or videos during class or of written examinations, and distributing such materials, is prohibited.

Student Well-being

University life can sometimes be challenging, and seeking support during this process is entirely natural. The Bahçeşehir University Psychological Counseling and Guidance Center (PCGC) provides support to students in their personal, social, and academic development. For PCGC services, please visit: <https://bau.edu.tr/content/17144-psychological-counseling-and-guidance-center>

11. Learning Resources

Textbook: "Numerical Analysis" by Faires & Burden, 2011.

Lecture notes and Lab notes will be uploaded to the Resources folder on Microsoft Teams.

More books in the Library:

- Stephen C. Chapra, Applied Numerical Methods W/MATLAB: for Engineers & Scientists, 3rd Edition, McGrawHill.
- Applied Numerical Methods Using MATLAB, Won Young Yang, Wenwu Cao, Tae-Sang Chung, John Morris.
- John H. Mathews and Kurtis D. Fink, Numerical Methods Using MATLAB, Pearson, 2004. ISBN 0-13-191178-3.

Reference resources for MATLAB:

- Introduction to MATLAB from Rice University: <http://www.cs.bilkent.edu.tr/~tugrul/CS471/unix17.pdf>
- MATLAB Primer by Kermit Sigmon: <http://www.cs.bilkent.edu.tr/~tugrul/CS471/matlab-primer.pdf>
- MATLAB on Athena, MIT: <http://web.mit.edu/afs/.athena/astaff/project/logos/olh/Math/Matlab/Matlab.html>
- More on MATLAB: <http://www.cse.uiuc.edu/heath/scicomp/matlab.html>
- Free scientific portal for Matlab: <http://www.mathtools.net>

12. Weekly Learning Plan

Week	Course Learning Outcome No	Topic	Learning Activities	Before Class	Assessment
1	1	Introduction to numerical analysis: kinds of problems we solve; Error analysis, round-off and truncation errors, Taylor theorem; Loss of significance; O notation.	Face-to-face lecture; solve self-study problems.	Read Lecture Note 1.	
2	2	Bisection and False-Position methods for the solution of nonlinear algebraic equations.	Face-to-face lecture; solve self-study problems.	Read the first part of Lecture Note 2.	
3	2	Fixed point iteration and Newton-Raphson methods for the solution of nonlinear algebraic equations.	Face-to-face lecture; solve self-study problems.	Read the relevant lecture note.	Quiz 1 (date announced on Teams)
4	3, 9	Solution of linear systems of equations; Upper-triangular linear systems; Gaussian elimination method with pivoting; LU factorization method with pivoting.	Face-to-face lecture; MATLAB problem solving; self-study problems.	Read Lecture Note 4.	MATLAB Quiz
5	3	Jacobi and Gauss-Seidel iterative methods for the solution of linear systems.	Face-to-face lecture; solve self-study problems.	Read Lecture Note 6.	
6	4, 9	Introduction to Newton's method for the solution of nonlinear systems. Revision before the midterm exam.	Face-to-face lecture; MATLAB problem solving; self-study problems.	Read Lecture Note 7; revise for the midterm.	MATLAB Quiz
7	1-4	Midterm exam.	Midterm exam.	Review Weeks 1-6 lecture notes and self-study problems.	Midterm Exam

Week	Course Learning Outcome No	Topic	Learning Activities	Before Class	Assessment
8	4	Newton's method for the solution of nonlinear systems. Revision before the midterm exam.	Face-to-face lecture; solve self-study problems.	Read Lecture Note 7.	
9	5	Interpolation using Lagrange and Newton polynomials.	Face-to-face lecture; solve self-study problems.	Read Lecture Note 8.	MATLAB Quiz
10	6	Curve fitting using least-squares method; Linearization of nonlinear relationships.	Face-to-face lecture; solve self-study problems.	Read Lecture Note 9.	
11	7, 9	Eigenvalues and eigenvectors of a matrix; Power method for approximating the dominant eigenvalue of a matrix; Singular Value Decomposition.	Face-to-face lecture; MATLAB problem solving; self-study problems.	Read the first part of Lecture Note 10.	MATLAB Quiz
12	8, 9	Numerical Differentiation; Numerical Integration Methods: Trapezoidal Rule and Simpson's Rule; Discrete Fourier Transform.	Face-to-face lecture; MATLAB problem solving; self-study problems.	Read Lecture Notes 11 and 12.	
13					
14					

13. Student Success Guide

- Read all weekly course materials and follow Microsoft Teams announcements regularly.
- Attend classes actively every week and participate positively in classroom activities and discussions.
- Complete the self-study exercises in the lecture notes and practice numerical methods in MATLAB during problem-solving/lab work.
- Use e-mail for questions outside class hours and seek clarification early when a topic or method is unclear.
- If a health issue or personal problem affects an examination, submit the required medical report to the department secretary; make-up examinations are governed by BAU regulations.

Date	Prepared by
21.09.2026	Lavdie Rada Ülgen